

Wmt Forecasting :Problem Understanding

1. Problem Statement

Forecast **daily unit sales** of Walmart retail products, at the item × store level, for **28 consecutive days into the future**.

- Covers **3 US states**: California (CA), Texas (TX), Wisconsin (WI)
- **10 stores, 3 product categories, 7 departments, 3,049 unique items**
- Bottom-level series: **30,490** (item × store combinations)
- Predictions are scored across **12 hierarchical aggregation levels** simultaneously

The task is a multi-horizon, hierarchical, point-forecast problem. It is notable because many series exhibit **intermittent demand** (sporadic sales, lots of zeros).

2. Data

2.1 Files (450 MB total)

File	Description	Key Columns
<code>sales_train_validation.csv</code>	Daily unit sales, d_1 → d_1913 (~5.4 years)	<code>id</code> , <code>item_id</code> , <code>dept_id</code> , <code>cat_id</code> , <code>store_id</code> , <code>state_id</code> , <code>d_1</code> ... <code>d_1913</code>
<code>sales_train_evaluation.csv</code>	Daily unit sales, d_1 → d_1941 (includes public LB ground truth)	Same as above through <code>d_1941</code>
<code>calendar.csv</code>	Date metadata for each day index	<code>date</code> , <code>wm_yr_wk</code> , <code>weekday</code> , <code>wday</code> , <code>month</code> , <code>year</code> , <code>d</code> , <code>event_name_1/2</code> , <code>event_type_1/2</code> , <code>snap_CA/TX/WI</code>
<code>sell_prices.csv</code>	Weekly sell price per item per store	<code>store_id</code> , <code>item_id</code> , <code>wm_yr_wk</code> , <code>sell_price</code>
<code>sample_submission.csv</code>	Correct submission format	<code>id</code> , <code>F1</code> ... <code>F28</code>

2.2 Data Scale

Dimension	Value
Days of history	1, 941 ($\approx$ 5.4 years, Jan 2011 – May 2016)
Bottom-level series	30,490
Forecast horizon	28 days
Total predictions per submission	60,980 rows $\times$ 28 = 1, 707, 440 values

2.3 Calendar Features

- **SNAP days:** Supplemental Nutrition Assistance Program benefit issuance days (state-specific), a major driver for food sales
- **Events:** Cultural/sporting/national events (Super Bowl, Valentine's Day, St. Patrick's Day, Easter, Mother's Day, Memorial Day, NBA Finals, Independence Day, Halloween, Thanksgiving, Christmas, etc.)
- **Event types:** Sporting, Cultural, National, Religious

2.4 Hierarchy Structure

Level	Description	Series count
1	Total (all items, all stores)	1
2	State	3
3	Store	10
4	Category (FOODS, HOBBIES, HOUSEHOLD)	3
5	Department (FOODS_1/2/3, etc.)	7
6	State $\times$ Category	9
7	State $\times$ Department	21
8	Store $\times$ Category	30
9	Store $\times$ Department	70
10	Item	3,049
11	Item $\times$ State	9,147
12	Item $\times$ Store $\leftarrow$ bottom level	30,490
Total series across all 12 levels		42,840

3. Metric: WRMSSE

The evaluation metric is the **Weighted Root Mean Squared Scaled Error (WRMSSE)**.

3.1 RMSSE (per series)

For a single aggregated series i with training observations $Y_1, \dots, Y_n$ and forecasts $\hat{Y}_{n+1}, \dots, \hat{Y}_{n+h}$ (where $h = 28$):

$$\text{RMSSE}_i = \sqrt{\frac{\sum_{t=n+1}^{n+h} (Y_{i,t} - \hat{Y}_{i,t})^2}{\sum_{t=2}^{n^*} (Y_{i,t} - Y_{i,t-1})^2}}$$

where:

- **Numerator:** Mean squared forecast error over the 28-day horizon
- **Denominator:** Mean squared one-step naive forecast error over the *active* training period (starting from the first non-zero demand observation, i.e. $n^* \leq n$)
- The denominator is the in-sample scaling factor — it makes the metric scale-independent and comparable across series of different magnitudes

Interpretation: RMSSE = 1.0 means the model performs exactly as well as a naive (random walk) baseline. Lower is better.

3.2 Weights (per level)

Within each aggregation level l , each series i receives a weight proportional to its **dollar-value revenue** over the last 28 days of training:

$$w_{i,l} = \frac{\sum_{t=n-27}^n p_{i,t} \cdot Y_{i,t}}{\sum_{i \in S_l} \sum_{t=n-27}^n p_{i,t} \cdot Y_{i,t}}$$

where $p_{i,t}$ is the sell price and $Y_{i,t}$ is the unit sales on day t .

Series that sell more (in dollar terms) get higher weight — the metric penalises errors on high-revenue items more.

3.3 WRMSSE (final score)

$$\text{WRMSSE} = \frac{1}{12} \sum_{l=1}^{12} \underbrace{\sum_{i \in S_l} w_{i,l}}_{\text{weighted RMSSE for level } l}$$

The 12 level scores are **equally weighted** in the final average.

3.4 Metric Implications

Implication	Why it matters
Score is scale-free	Can compare across items at very different sales volumes
Revenue-weighted	Models must perform well on high-value items (FOODS > HOBBIES in weight)

Hierarchy-averaged	Can't just optimise bottom level — errors aggregate up
Intermittency	If a series has only zeros, it is excluded from denominator calculation (no division by zero)
Lower bound	Theoretically > 0 ; a perfect forecast scores 0

4. Submission Format

```
id, F1, F2, ..., F28
HOBBIES_1_001_CA_1_validation, 0, 1, ..., 2
HOBBIES_1_001_CA_1_evaluation, 3, 2, ..., 7
```

- **60,980 rows** total: 30,490 `_validation` rows + 30,490 `_evaluation` rows
- `validation` → forecasts for **d_1914 to d_1941** (scored on public leaderboard)
- `evaluation` → forecasts for **d_1942 to d_1969** (scored on private leaderboard)

5. Key Modelling Challenges

Challenge	Description
Intermittent demand	Many bottom-level series have $>50\%$ zero days; Poisson / Tweedie loss functions more appropriate than Gaussian
Scale	30,490 series; global/cross-sectional models (e.g. LightGBM, CatBoost) are practical; per-series models are infeasible
Hierarchy consistency	Predictions should ideally respect sum-of-parts (reconciliation) — though not strictly required by the metric
Pricing effects	Price changes affect demand non-linearly; lag features of price are important
Calendar effects	Day of week, month, SNAP days, and holiday effects have strong predictive power
Cold starts	Some items are only introduced partway through the history; the active-period denominator in RMSSE accounts for this
Revenue weighting	FOODS items are far more revenue-weighted than HOBBIES or HOUSEHOLD — worth tuning model on FOODS more

6. Modelling Approaches (from competition)

Top-performing solutions used:

1. **LightGBM (GBDT)** — most dominant; global model trained over all series with lag features, rolling statistics, calendar encodings
2. **Neural networks** — DeepAR, WaveNet-style 1D CNNs, Temporal Fusion Transformer
3. **Ensemble** — weighted combination of GBDT + NN + statistical models (ETS, ARIMA, Croston)
4. **Feature engineering** — lag-1/7/14/28 sales, rolling mean/std windows (7/14/28/56/180 days), price features (relative price vs. category average), event flags, SNAP flags

7. Data Download

```
# Requires: pip install kaggle + ~/.kaggle/kaggle.json + accepted competition rule
kaggle competitions download -c m5-forecasting-accuracy -p data/
cd data && unzip m5-forecasting-accuracy.zip
```

8. Quick Reference: Column Glossary

sales_train_*.csv

Column	Description
id	Unique identifier: {item_id}_{store_id}_{split}
item_id	e.g. HOBBIES_1_001
dept_id	Department: HOBBIES_1 , FOODS_2 , etc.
cat_id	Category: HOBBIES , FOODS , HOUSEHOLD
store_id	e.g. CA_1 , TX_2
state_id	CA , TX , WI
d_1 ... d_N	Daily unit sales (integer)

calendar.csv

Column	Description
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d	Day index (d_1 ... d_1969)
date	Calendar date (YYYY-MM-DD)
wm_yr_wk	Walmart fiscal year-week (links to sell_prices.csv)
weekday / wday	Day name / day number (1=Saturday)
event_name_1/2	Name of event if any
event_type_1/2	Type: Sporting / Cultural / National / Religious
snap_CA/TX/WI	Binary: 1 if SNAP benefits issued in that state that day

sell_prices.csv

Column	Description
store_id	Store identifier
item_id	Item identifier
wm_yr_wk	Walmart fiscal week
sell_price	Selling price (USD) for that week